



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



NASA Core Flight System Health and Safety Vulnerability Daily Space Systems Review Update

Threat Report

Classification	TLP:CLEAR
Published	2026-08-14
Version	1.0
Author	Lost Boys Cyber
Report Type	Threat Report

TLP:CLEAR | Lost Boys Cyber | NASA Core Flight System Health and Safety Vulnerability Daily Space Systems Review Update - Threat Report

Published: 2026-08-14 | TLP:CLEAR | Version: 1.0 | Author: Lost Boys Cyber

AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

Table of Contents

1. Executive Summary
2. Intelligence Requirements
3. Source Review & Confidence
4. Threat Activity Overview
5. Affected Sectors and Exposure
6. Aviation and Aerospace Sector Impact
7. Tactics, Techniques, and Procedures
8. Infrastructure and Indicators
9. Detection Engineering
 - 9.1. Detection Summary
 - 9.2. Sigma / Detection Content
 - 9.3. Hunt Query
10. Threat Hunting
 - 10.1. Hunt Hypothesis
 - 10.2. Hunt Query
11. Risk Assessment
12. Recommended Actions
13. References

NASA Core Flight System Health and Safety Vulnerability Daily Space Systems Review Update

1. Executive Summary

NASA Core Flight System Health and Safety vulnerabilities remain relevant for aerospace defenders because CISA ICS advisories describe denial-of-service conditions affecting cFS HS application versions. Space-system operators should verify component exposure, patch level, and mission-specific fault handling.

This daily review prioritizes practical defender action over attribution certainty. Source depth was expanded to the last seven days where needed to preserve high-confidence, multi-source reporting rather than fabricate same-day claims.

2. Intelligence Requirements

Question	Current Answer	Confidence
What changed for defenders today?	Space-system software vulnerabilities require sector-specific exposure review, not generic CVE handling.	Medium
Who is most exposed?	Satellite operators, research missions, aerospace labs, and integrators using cFS components.	Medium
What immediate action should analysts take?	Inventory cFS deployments, patch/test HS application updates, and review fault-response procedures.	High

3. Source Review & Confidence

Source	Contribution	Confidence
CISA ICSA-26-211-06 NASA cFS HS advisory	Primary ICS advisory	High
CISA ICSA-26-197-03 NASA cFS HS advisory	Related cFS vulnerability advisory	High
NVD CVE-2026-15352	Vulnerability metadata context	High

4. Threat Activity Overview

The issue is not presented here as active exploitation. Its importance is operational: flight software and mission-support environments require controlled patch testing, redundancy assessment, and clear mission-impact analysis.

5. Affected Sectors and Exposure

Sector / Environment	Exposure	Analyst Priority
Enterprise identity and endpoint estates	Credential abuse, script execution, and malware/ransomware staging.	Review high-risk sign-ins, new persistence, and anomalous script activity.
Cloud and SaaS administration	Token theft, OAuth abuse, risky automation, and data exposure.	Audit consent grants, service principals, and admin actions.
Managed service / supply chain	Shared access paths amplify impact.	Validate supplier controls and inbound trust

relationships.

6. Aviation and Aerospace Sector Impact

Segment	Operational Relevance	Exposure Path	Defender Action
Spacecraft software, satellite operators, and aerospace engineering teams using cFS-derived components	Denial-of-service in flight software support components can affect mission resilience and test environments.	Identity, supplier access, exposed services, SaaS, OT/IT integration, or satellite/ground-system dependency.	Validate segmentation, vendor access, MFA posture, backups, and incident communications.
Third-party ecosystem	Aviation depends on MRO, airport, reservation, weather, satellite, and defense supply-chain partners.	Compromised managed service, shared platform, or supplier credential.	Require supplier attestations for patching, logging, and notification SLAs.

7. Tactics, Techniques, and Procedures

Tactic / Phase	Observed Technique	Evidence
Initial Access	Phishing, exposed service exploitation, or supplier compromise	Source reporting describes active threat activity or sector-relevant risk.
Execution	Command/script execution and automation abuse	Daily detection focus includes process and cloud-control-plane telemetry.
Persistence / Defense Evasion	Token/OAuth misuse, service changes, tooling overlap	Hunt for abnormal long-lived access and administrative changes.
Impact	Data theft, ransomware/extortion, or operational disruption	Reported campaigns and sector risk concentrate on business disruption.

8. Infrastructure and Indicators

Indicator / Infrastructure	Type	Operational Use
Public sources did not provide stable atomic IOCs for all environments	Limitation	Use behavior-led detections and enrich with local/vetted vendor IOCs.
NASA cFS	Keyword / analytic pivot	Use in triage notes, EDR search, and CTI tagging; not an IOC by itself.

9. Detection Engineering

9.1. Detection Summary

Signal	Telemetry Source	Analytic Goal
Suspicious script or command execution after lure/access event	EDR process telemetry	Identify payload staging, reconnaissance, or automation abuse.
Unusual outbound destination or direct-to-IP access	Network/EDR DNS and connection telemetry	Surface malware C2, staging, or infrastructure bypassing DNS controls.
New OAuth grant, token use, or admin consent anomaly	Cloud identity logs	Detect identity-centric intrusion and persistence.

9.2. Sigma / Detection Content

```

title: Daily Threat Activity Signal - NASA cFS
id: daily-20260814-daily-threat-activity-signal-nasa-cfs
status: experimental
description: Analyst-facing daily-review detection seed for Daily Threat Activity Signal - NASA cFS.
references:
  - https://github.com/lost0x01/lost-boys-cyber-analyst-kit
author: Lost Boys Cyber
date: 2026/08/14
logsource:
  category: process_creation
detection:
  selection_keywords:
    Keyword1: "NASA cFS"
    Keyword2: "ransomware"
    Keyword3: "token"
  condition: selection_keywords
falsepositives:
  - Administrative scripts or vulnerability-scanner probes using the same product strings.
level: medium

```

9.3. Hunt Query

```

let lookback = 14d;
let terms = dynamic(["NASA cFS", "ransomware", "oauth", "token"]);
DeviceNetworkEvents
| where Timestamp > ago(lookback)
| where RemoteUrl has_any (terms) or InitiatingProcessCommandLine has_any (terms)
| summarize FirstSeen=min(Timestamp), LastSeen=max(Timestamp), Hosts=dcount(DeviceName),
Samples=make_set(InitiatingProcessCommandLine, 5) by RemoteUrl, RemoteIP, InitiatingProcessFileName
| order by LastSeen desc

```

10. Threat Hunting

10.1. Hunt Hypothesis

Relevant activity will present as behavior chains rather than a single static IOC: lure or exploit trigger, script/process execution, identity or token changes, outbound staging, and data-access anomalies.

10.2. Hunt Query

```

let lookback = 14d;
let pivots = dynamic(["NASA cFS", "oauth", "token", "ransomware", "extortion"]);
DeviceProcessEvents
| where Timestamp > ago(lookback)
| where ProcessCommandLine has_any (pivots) or InitiatingProcessCommandLine has_any (pivots)
| summarize FirstSeen=min(Timestamp), LastSeen=max(Timestamp), Hosts=dcount(DeviceName),
Examples=make_set(ProcessCommandLine, 5) by FileName, InitiatingProcessFileName
| order by LastSeen desc

```

11. Risk Assessment

Risk	Likelihood	Impact	Notes
Identity-led intrusion	Medium	High	Current reporting continues to show credential/token misuse as a durable access path.
Ransomware/extortion	Medium	High	Impact depends on segmentation, backup readiness, and supplier blast radius.
Operational disruption	Medium	High	Highest for aviation/aerospace,

			healthcare, manufacturing, and managed-service dependencies.
--	--	--	--

12. Recommended Actions

Priority	Action
P0	Review whether the organization uses affected products/services or matches the reported exposure pattern.
P1	Run the included hunts, tune with local IOCs, and validate alerts with identity/network context.
P1	Confirm MFA, conditional access, vendor access, backups, and incident communications for affected business owners.
P2	Add source URLs and report metadata to CTI tracking/OpenCTI for recurring review.

13. References

- [1] CISA ICSA-26-211-06 NASA cFS HS advisory: <https://www.cisa.gov/news-events/ics-advisories/icsa-26-211-06>
- [2] CISA ICSA-26-197-03 NASA cFS HS advisory: <https://www.cisa.gov/news-events/ics-advisories/icsa-26-197-03>
- [3] NVD CVE-2026-15352: <https://nvd.nist.gov/vuln/detail/CVE-2026-15352>